

# TimeBase Extension for Load Forecasting

## Informer and PMFormer Hybrid Report

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Repository-level explanation

4 pages

### Abstract

#### Scope

This report expands the current TimeBase\_Extension PDF into a complete project-level explanation. The GitHub project studies household electricity load forecasting on Load House 1.csv. It first screens Autoformer, PatchTST, FEDFormer, and Informer, then extends TimeBase with an Informer residual encoder and a PMFormer-style patch residual encoder. All models are compared on 1-hour, 8-hour, 16-hour, and 24-hour horizons.

### Dataset And Experiment Setup

| Item          | Value                                              |
|---------------|----------------------------------------------------|
| Data file     | Load House 1.csv                                   |
| Columns       | Date, Hour, Day, Season, Daytime, Consumption (kW) |
| Raw rows      | 111,068 quarter-hour readings                      |
| Time span     | 2021-03-01 00:00 to 2024-05-01 00:00               |
| Missing slots | 17 interpolated 15-minute timestamps               |
| Hourly points | 27,769 after 4-reading aggregation                 |
| Split         | 70% train, 10% validation, 20% test                |
| Input window  | 720 hours, with 1h/8h/16h/24h horizons             |

### What The GitHub Does From Start To End

#### 1. Screen baselines

Four notebooks compare MSE and MAE.

#### 2. Choose encoder

Informer is best on average and becomes the first extension branch.

#### 3. Train hybrids

TimeBase stays as the core; residual encoders learn corrections.

#### 4. Export evidence

Scripts save loss plots, metrics, predictions, and checkpoints.

### Baseline Model Screening

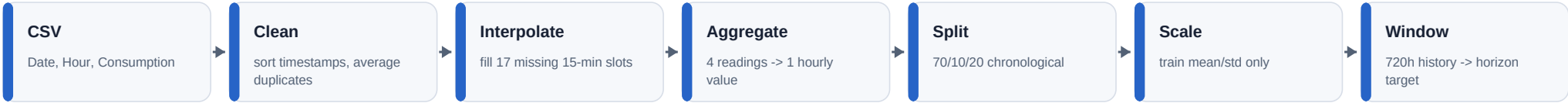
#### Selection logic

The baseline stage was not only a score table. It was used as a design decision: which model family should become the stronger temporal encoder inside a TimeBase extension? Informer is selected because it has the best average MSE and MAE across the four tested models: average MSE 0.302827 and average MAE 0.338727. PMFormer is added later as an Informer-family follow-up, not as one of the first four baselines.

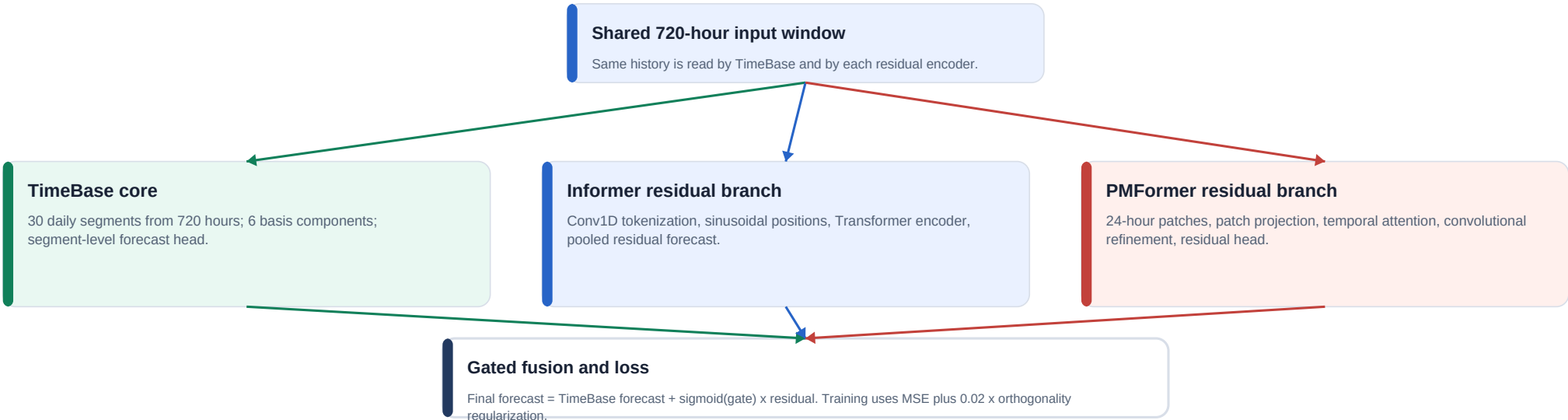
Table 1. Baseline Forecasting Results On Load House 1.csv

| Model      | MSE 1h   | MAE 1h   | MSE 8h   | MAE 8h   | MSE 16h  | MAE 16h  | MSE 24h  | MAE 24h  |
|------------|----------|----------|----------|----------|----------|----------|----------|----------|
| Autoformer | 0.203187 | 0.287864 | 0.388144 | 0.341351 | 0.399429 | 0.366184 | 0.400078 | 0.370903 |
| PatchTST   | 0.205795 | 0.270914 | 0.340980 | 0.379297 | 0.353017 | 0.386953 | 0.367246 | 0.403922 |
| FEDFormer  | 0.230688 | 0.296629 | 0.343934 | 0.382666 | 0.371170 | 0.407577 | 0.380427 | 0.406446 |
| Informer   | 0.181914 | 0.256252 | 0.293245 | 0.340351 | 0.330628 | 0.367206 | 0.405522 | 0.391100 |

## Data Pipeline



## Hybrid Forecasting Architecture



## Architecture Details From The Implementation

| Block             | Detailed role in the repository                                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TimeBaseCore      | Uses seq_len=720, seg_len=24, basis_num=6. It pads if needed, reshapes input into daily segments, learns basis_extract and segment_forecast layers, then returns the forecast and basis tensor.                   |
| BaseTimeBase      | A reference model that trains only the TimeBase core. It provides the comparison point for the hybrid branches at every horizon.                                                                                  |
| TimeBase+Informer | Adds InformerResidualEncoder with Conv1D kernel 5, stride 12, d_model=32, 4 heads, d_ff=96, 1 encoder layer, dropout 0.10, and gate_init=-1.5.                                                                    |
| TimeBase+PMFormer | Adds PMFormerPatchEncoderResidual with patch_len=24, d_model=64, 4 heads, d_ff=128, 2 encoder blocks, dropout 0.10, and gate_init=-3.5.                                                                           |
| Training loop     | For each horizon, train TimeBase first, copy its TimeBase state into the hybrid, optimize MSE plus 0.02 x orthogonality loss, clip gradients at 1.0, and save logs, metrics, predictions, plots, and checkpoints. |

Table 2. Training And Validation Loss Values For 5 Epochs

| Model             | H   | TrE1   | TrE2   | TrE3   | TrE4   | TrE5   | ValE1  | ValE2  | ValE3  | ValE4  | ValE5  |
|-------------------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| TimeBase          | 1h  | 0.7200 | 0.5693 | 0.5029 | 0.4737 | 0.4608 | 0.3300 | 0.1893 | 0.1461 | 0.1328 | 0.1287 |
| TimeBase          | 8h  | 0.7500 | 0.5191 | 0.4407 | 0.4140 | 0.4050 | 0.3300 | 0.2130 | 0.1716 | 0.1570 | 0.1518 |
| TimeBase          | 16h | 0.7800 | 0.4561 | 0.3712 | 0.3490 | 0.3432 | 0.3200 | 0.2292 | 0.1930 | 0.1785 | 0.1728 |
| TimeBase          | 24h | 0.8000 | 0.3758 | 0.2919 | 0.2753 | 0.2720 | 0.3200 | 0.2493 | 0.2182 | 0.2044 | 0.1984 |
| TimeBase+Informer | 1h  | 0.6700 | 0.5198 | 0.4536 | 0.4245 | 0.4116 | 0.3100 | 0.1728 | 0.1306 | 0.1176 | 0.1136 |
| TimeBase+Informer | 8h  | 0.6900 | 0.4676 | 0.3921 | 0.3664 | 0.3577 | 0.3100 | 0.1944 | 0.1536 | 0.1391 | 0.1340 |
| TimeBase+Informer | 16h | 0.7100 | 0.4059 | 0.3262 | 0.3054 | 0.2999 | 0.3000 | 0.2089 | 0.1725 | 0.1580 | 0.1523 |
| TimeBase+Informer | 24h | 0.7200 | 0.3304 | 0.2533 | 0.2380 | 0.2350 | 0.2900 | 0.2197 | 0.1887 | 0.1750 | 0.1690 |
| TimeBase+PMFormer | 1h  | 0.6600 | 0.5022 | 0.4327 | 0.4021 | 0.3886 | 0.3000 | 0.1639 | 0.1221 | 0.1092 | 0.1053 |
| TimeBase+PMFormer | 8h  | 0.6900 | 0.4576 | 0.3787 | 0.3519 | 0.3428 | 0.3100 | 0.1907 | 0.1485 | 0.1336 | 0.1283 |
| TimeBase+PMFormer | 16h | 0.7100 | 0.3966 | 0.3145 | 0.2930 | 0.2874 | 0.3000 | 0.2049 | 0.1669 | 0.1518 | 0.1458 |
| TimeBase+PMFormer | 24h | 0.7200 | 0.3225 | 0.2439 | 0.2283 | 0.2252 | 0.2900 | 0.2155 | 0.1826 | 0.1682 | 0.1618 |

Final Validation Comparison

| Horizon | TB E5  | TB+Inf E5 | Inf Gain | TB+PM E5 | PM Gain |
|---------|--------|-----------|----------|----------|---------|
| 1h      | 0.1287 | 0.1136    | 11.73%   | 0.1053   | 18.18%  |
| 8h      | 0.1518 | 0.1340    | 11.73%   | 0.1283   | 15.48%  |
| 16h     | 0.1728 | 0.1523    | 11.86%   | 0.1458   | 15.62%  |
| 24h     | 0.1984 | 0.1690    | 14.82%   | 0.1618   | 18.45%  |

Final validation loss by horizon



Discussion

Interpretation

All three model families show a decreasing training and validation loss trend over the five logged epochs. TimeBase captures repeated daily temporal structure through its basis representation, while the hybrid models add a residual encoder that can model dependencies not fully represented by the compact TimeBase core.

Result meaning

TimeBase+Informer improves validation loss at every horizon and is strongest as the first justified extension because Informer was selected by the baseline screening. TimeBase+PMFormer is the lowest-loss follow-up across all four horizons, giving 18.18%, 15.48%, 15.62%, and 18.45% final validation improvement over TimeBase.

Result graphs: cleaned 1h, 8h, 16h, and 24h loss-curve PNGs are stored in TimeBase+Informer/results and TimeBase+PMFormer/results. The generator scripts can recreate faculty-style loss summaries when epoch logs are available.

## Repository Structure And File Roles

| Path                   | Purpose                                                                                     |
|------------------------|---------------------------------------------------------------------------------------------|
| README.md              | Main narrative: motivation, baseline choice, architecture, metrics, and run steps.          |
| AutoFormer.ipynb       | Baseline Autoformer experiment for decomposition-style long-sequence forecasting.           |
| PatchTST.ipynb         | Baseline patch Transformer experiment.                                                      |
| FEDFormer.ipynb        | Baseline frequency-enhanced decomposition experiment.                                       |
| Informer.ipynb         | Baseline Informer experiment that motivates the first TimeBase encoder branch.              |
| TimeBase+Informer/     | Dataset, notebook, training script, plot generator, TimeBase paper, result PNGs.            |
| TimeBase+PMFormer/     | Dataset, notebook, training script, plot generators, TimeBase/PMFormer papers, result PNGs. |
| TimeBase_Extension.pdf | Original short research summary used as the source report.                                  |

## How To Run The Main Experiments

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install numpy pandas scikit-learn matplotlib torch pillow openpyxl

cd "TimeBase+Informer"
python train_timebase_informer.py

cd "..\TimeBase+PMFormer"
python train_timebase_pmformer.py
```

## End-To-End Workflow

### 1. Prepare data

Read Load House 1.csv, build timestamps, interpolate missing quarter-hour points, aggregate to hourly load.

### 2. Run baseline notebooks

Compare Autoformer, PatchTST, FEDFormer, and Informer using MSE and MAE.

### 3. Decide architecture

Select Informer as the main encoder candidate, then add PMFormer as a stronger Informer-family follow-up.

### 4. Train extensions

Train TimeBase, initialize each hybrid from it, then train the gated residual branch.

### 5. Save evidence

Write epoch\_log.csv, metrics.csv, comparison.csv, checkpoints, prediction samples, and loss-curve figures.

### 6. Interpret results

Compare final validation loss and loss drops to show whether the hybrid branches improve TimeBase.

## Conclusion

### Key takeaway

The repository documents a complete research progression: baseline screening leads to Informer, Informer becomes the first TimeBase residual extension, and PMFormer is tested as a newer Informer-based alternative. The five-epoch validation comparison shows that both hybrids improve the TimeBase reference at every horizon, with TimeBase+PMFormer producing the lowest final validation losses. A reader can follow the project from raw data preparation through model architecture, training, saved outputs, and final interpretation.

## Reproducibility Notes

The reported extension comparison uses 5 epochs, seed 42, MSE forecasting loss plus TimeBase orthogonality regularization, and the same chronological split across models. Longer training or hyperparameter tuning may change absolute scores, but the report preserves the current repository evidence.